

eSupplementary information

Rapid Screening of Urinary Tract Infection using Microfluidic Inertial-Impedance Cytometry

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Conflict of interest statement: The authors have declared that no conflict of interest exists.

Table S1 Clinical demographics of urine samples.

Sample	Study Number	Organism	Bacteria	Clinical count				Impedance count			
				RBC (cells/ μ L)	WBC (cells/ μ L)	EPI (cells/ μ L)	Total cells (WBC+EPI)	Measurement time (sec)	Dispensed volume (μ L)	Total count (cells)	cells/ μ L
1	794505	No Growth	Not Seen	30	2	0	2	177	236.0	679	2.9
2	463405	Escherichia coli	1+	65	225	4	229	167	222.7	45701	205.2
3	738815	No Growth	Not Seen	11	87	5	92	179	238.7	9920	41.6
4	857415	Klebsiella pneumoniae	Not Seen	11	5	5	10	184	245.3	7977	32.5
5	658915	No Growth	Not Seen	14	37	1	38	180	240.0	6940	28.9
6	895815	Klebsiella pneumoniae	Not Seen	1	1	4	5	190	253.3	487	1.9
7	187350	No Growth	Not Seen	6	14	3	17	183	244.0	503	2.1
8	715335	Enterococcus faecalis	3+	3	225	2	227	182	242.7	83115	342.5
9	409835	No Growth	Not Seen	5	6	3	9	175	233.3	720	3.1
10	495735	Escherichia coli	3+	13	31	31	62	170	226.7	8199	36.2
11	991935	No Growth	Not Seen	2	2	6	8	177	236.0	2916	12.4
12	768835	Enterococcus faecalis	Not Seen	112	25	1	26	169	225.3	3115	13.8
13	789835	Klebsiella pneumoniae	3+	12	225	4	229	184	245.3	90585	369.2
14	980345	No Growth	Not Seen	83	3	2	5	188	250.7	4399	17.5
15	271455	Enterococcus faecalis	1+	15	55	2	57	179	238.7	2826	11.8
16	454855	Escherichia coli	3+	2	69	3	72	203	270.7	2397	8.9
17	675955	No Growth	Not Seen	5	2	2	4	180	240.0	534	2.2
18	042565	No Growth	Not Seen	12	2	2	4	184	245.3	177	0.7
19	000465	No Growth	Not Seen	<1	1	2	3	180	240.0	1391	5.8
20	108165	Escherichia coli	3+	113	225	7	232	181	241.3	154538	640.4
21	371714	Escherichia coli	3+	6	225	8	233	181	241.3	42618	176.6
22	611814	No growth	Not Seen	12	2	2	4	178	237.3	586	2.5
23	617614	Escherichia coli	3+	>225	225	11	236	183	244.0	48066	197.0
24	800124	No growth	Not Seen	3	4	0	4	181	241.3	402	1.7
25	851814	No growth	Not Seen	13	32	7	39	181	241.3	9976	41.3
26	883614	Klebsiella pneumoniae	2+	34	22	2	24	192	256.0	2894	11.3
27	085244	No growth	Not Seen	10	2	10	12	183	244.0	1427	5.8
28	518144	Escherichia coli	3+	3	95	1	96	178	237.3	13748	57.9
29	546834	Klebsiella pneumoniae	Not Seen	9	71	23	94	177	236.0	10106	42.8
30	631544	No growth	1+	10	31	7	38	178	237.3	3780	15.9
31	808344	No growth	Not Seen	4	106	22	128	175	233.3	17689	75.8
32	823044	Escherichia coli	3+	222	225	12	237	177	236.0	148747	630.3
33	116864	No growth	Not Seen	2	2	3	5	177	236.0	338	1.4
34	125664	Escherichia coli	3+	13	89	18	107	129	172.0	10851	63.1
35	289664	Klebsiella pneumoniae	Not Seen	<1	3	11	14	183	244.0	1736	7.1
36	465074	No growth	Not Seen	50	2	7	9	179	238.7	932	3.9
37	620074	No growth	Not Seen	1	1	2	3	181	241.3	244	1.0
38	647664	Klebsiella pneumoniae	3+	18	225	22	247	194	258.7	106744	412.7
39	771174	No growth	Not Seen	26	1	1	2	177	236.0	228	1.0
40	913764	Escherichia coli	3+	10	225	5	230	182	242.7	126372	520.8

*RBC – Red blood cells, WBC – White blood cells, EPI – Epithelial cells

Table S2 Clinical demographics of neutrophils/bacteria spiked urine samples.

Sample Label	Donor	Condition	Urine amount (μL)	Neutrophil amount (μL)	E. coli amount (μL)
A	1	non-UTI	2000	-	-
B	2	UTI	1687	300	13
C	1	non-UTI	2000	-	-
D	1	UTI	1300	500	200
E	2	non-UTI	2000	-	-
F	3	UTI	1898	100	2
G	4	non-UTI	2000	-	-
H	4	UTI	1969.6	30	0.4
I	3	non-UTI	2000	-	-
J	1	UTI	1860	130	10

** Neutrophils stock was 1E+06 cells/mL and bacterial stock was 1E+08 CFU/mL*

***All samples were incubated in orbital shaker at 37 °C, 200 rpm, for 4 h before placing in 4 °C fridge overnight prior impedance measurement*

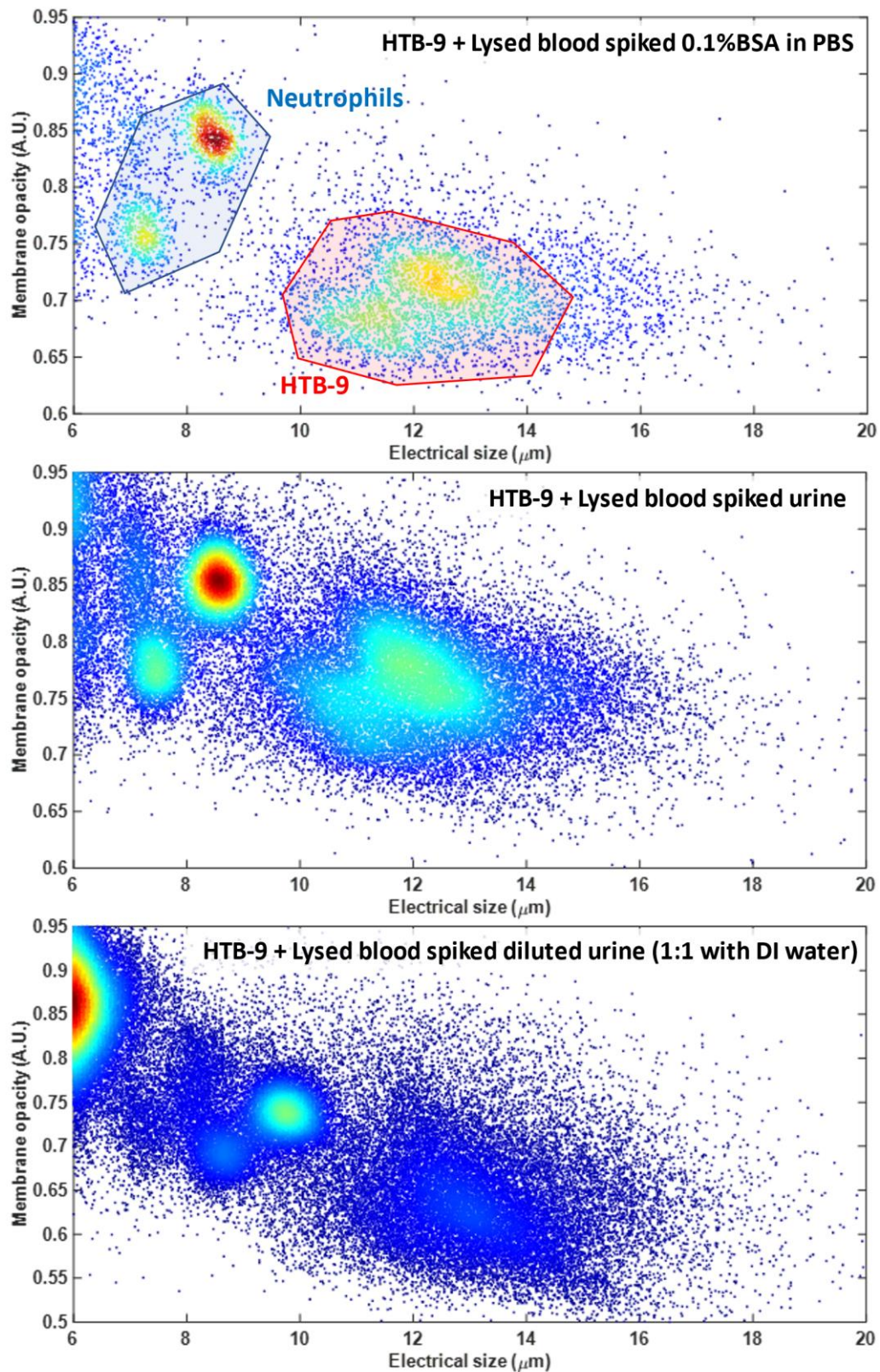


Figure S1 Impedance profiling of neutrophils (lysed blood) and HTB-9 in different mediums namely PBS (top), urine (center) and deionized water diluted urine (bottom). The measurements were done using single inlet inertial impedance cytometer to demonstrate device performance without buffer exchange. Dual clusters for each population were due to inertial focusing. Differences in electrical characteristics could be attributed to different medium properties.

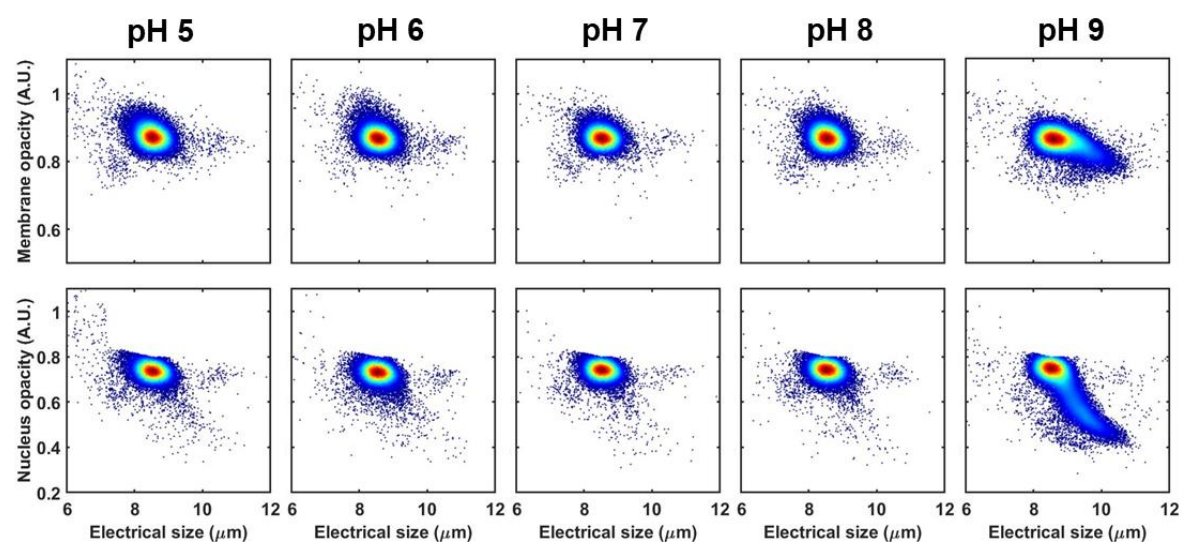


Figure S2 Impedance profiling of neutrophils in different pH.

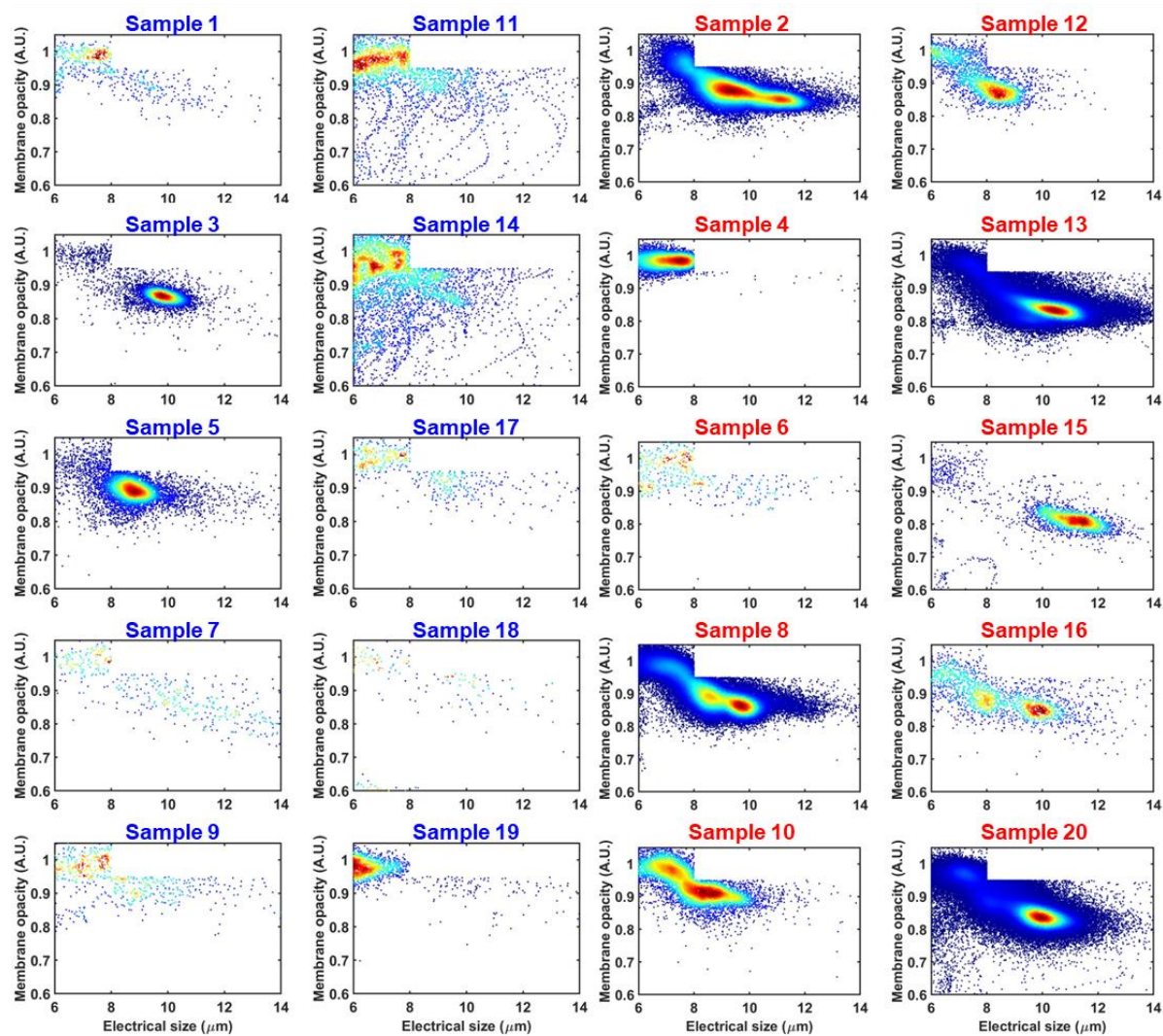


Figure S3 Dual frequency impedance profiling of urine samples (Sample 1-20)

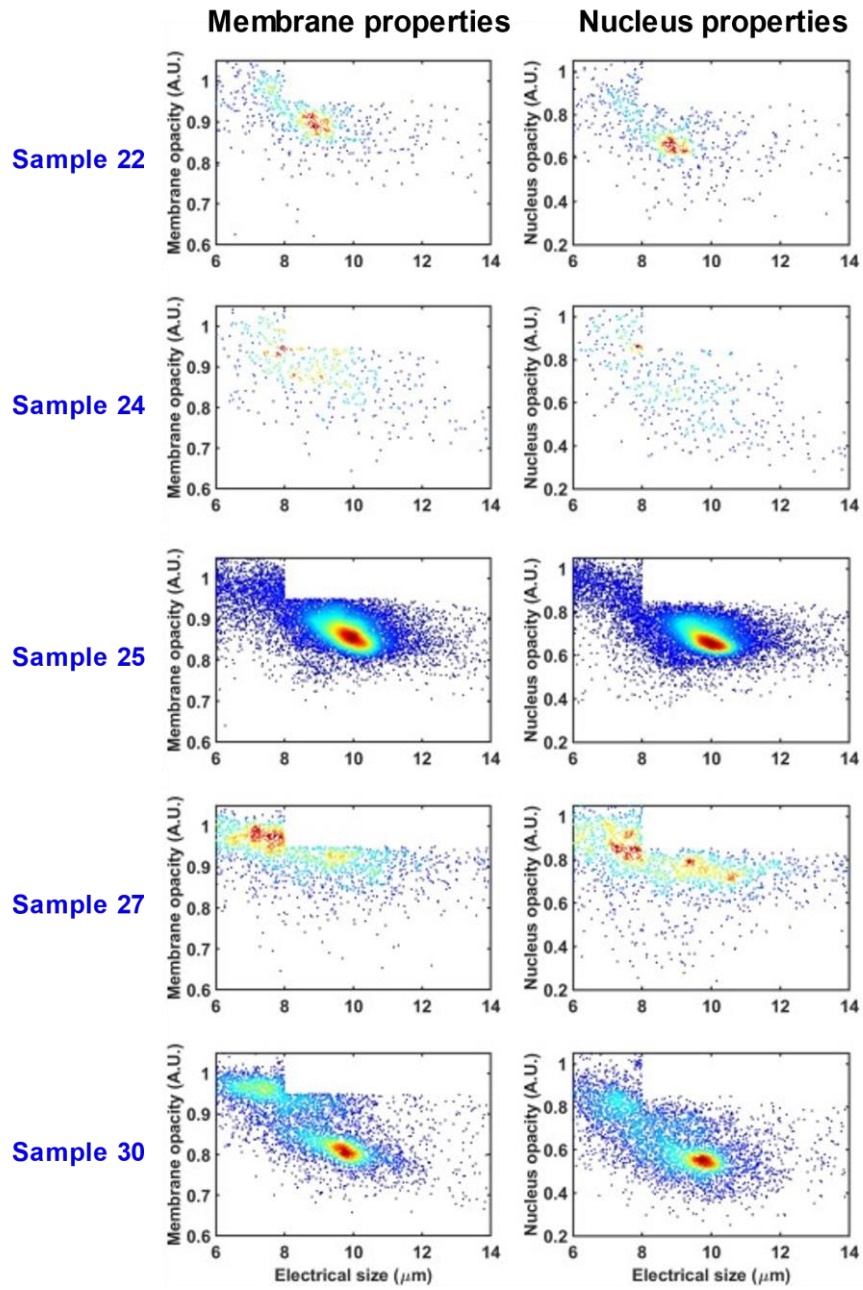


Figure S4 Triple frequency impedance profiling of non-UTI urine samples (sample 22, 24, 25, 27, 30)

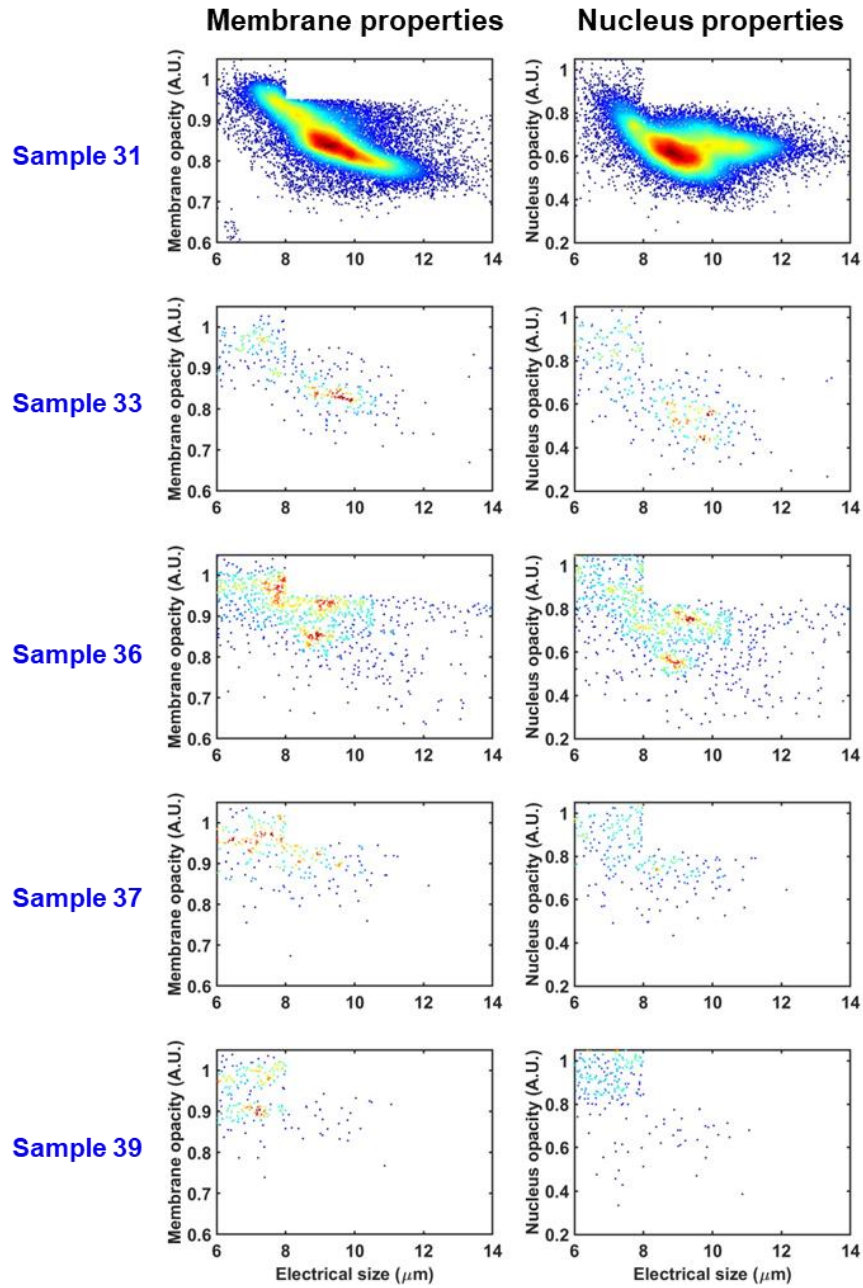


Figure S5 Triple frequency impedance profiling of non-UTI urine samples (sample 31, 33, 36, 37, 39)

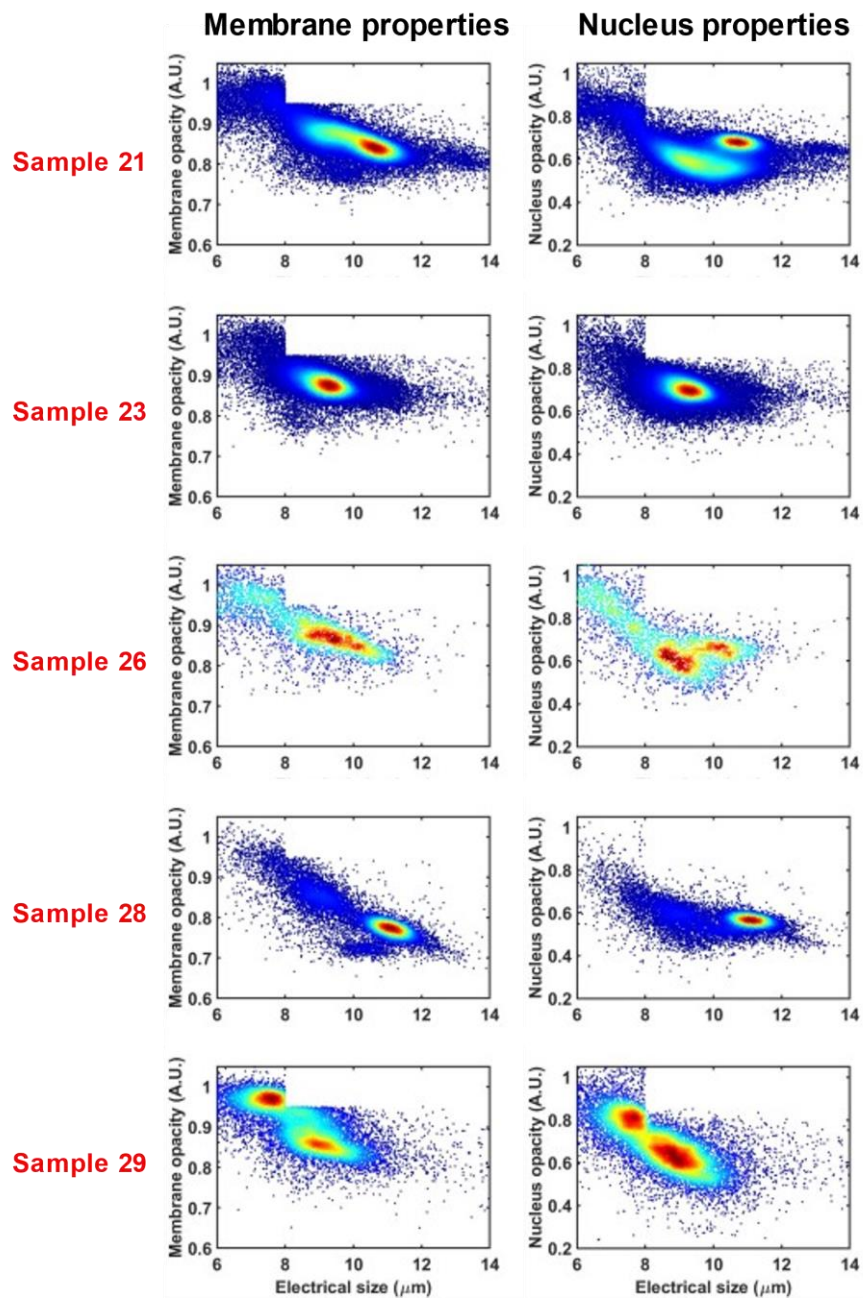


Figure S6 Triple frequency impedance profiling of UTI urine samples (sample 21, 23, 26, 28, 29)

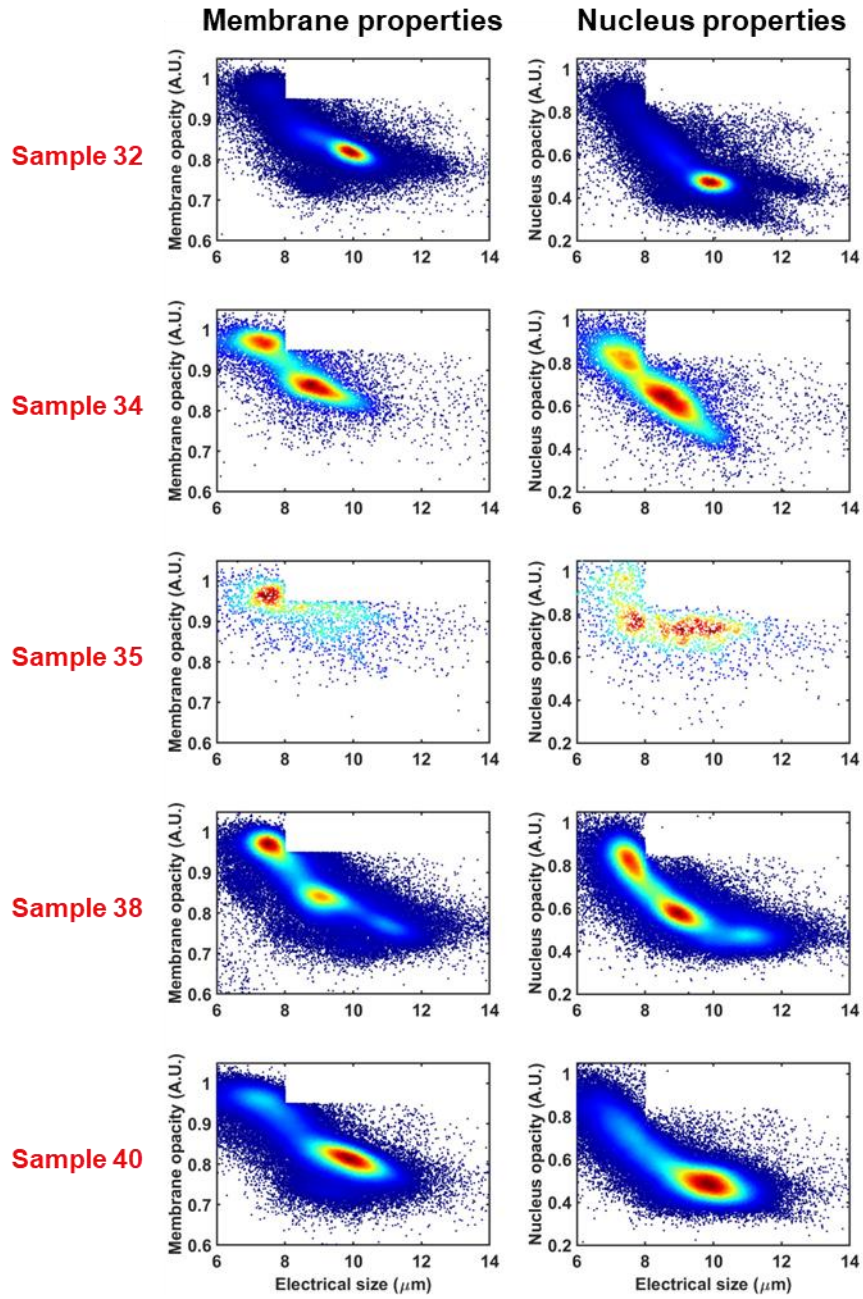


Figure S7 Triple frequency impedance profiling of UTI urine samples (sample 32, 34, 35, 38, 40)

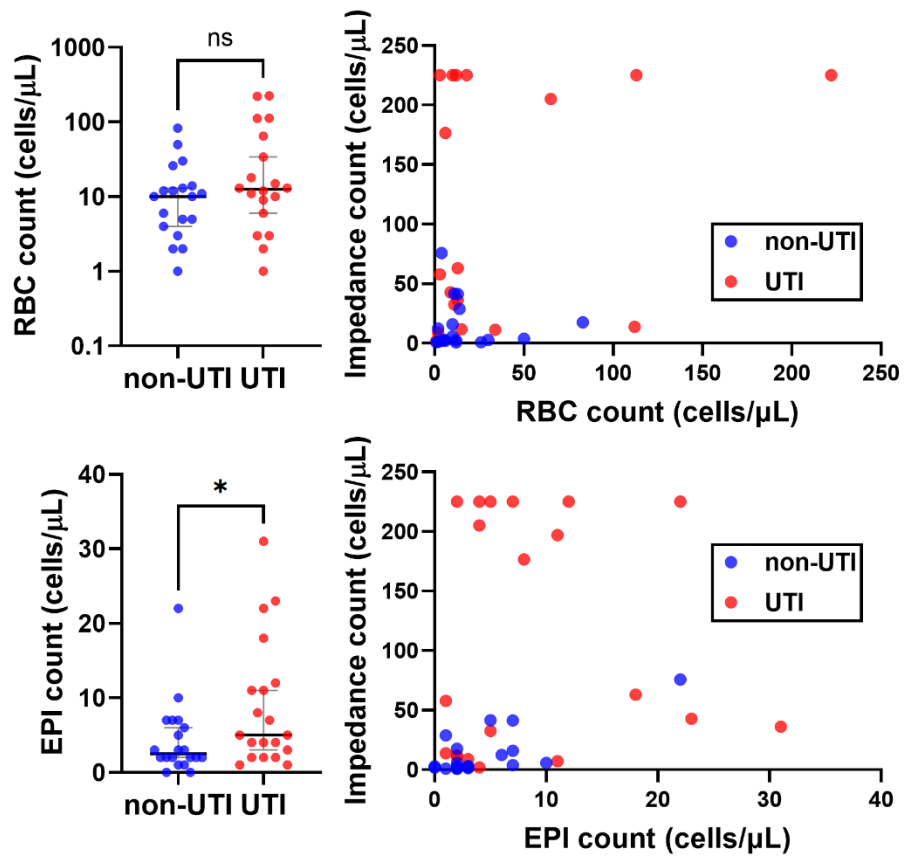


Figure S8 Clinical counts for red blood cells (RBC) (top) and epithelial cells (bottom) and their correlation with impedance count. Data are presented as median $\pm$ 95%CI. * $P \leq 0.05$.

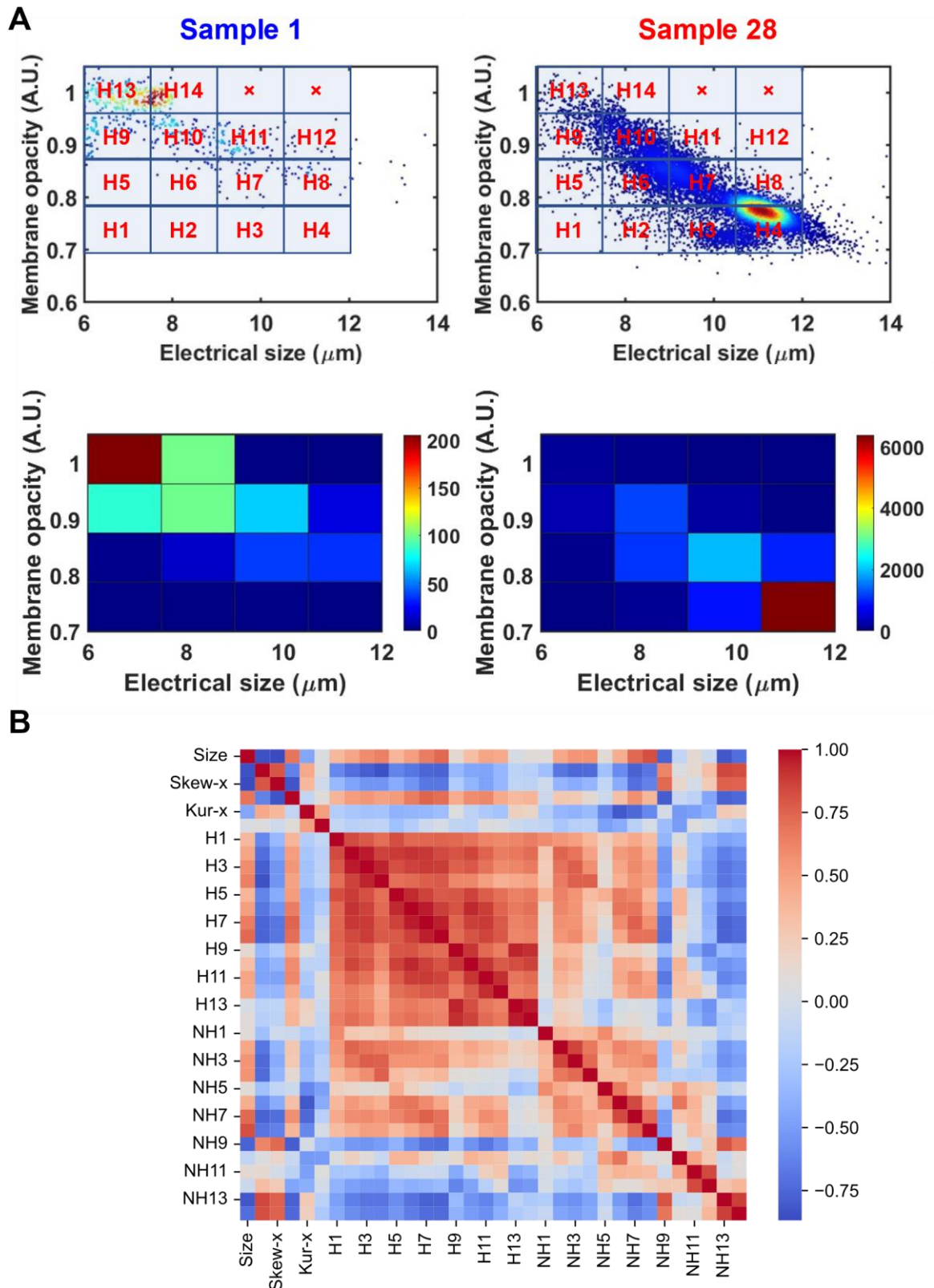


Figure S9 Feature definition and analysis. (A) bivariate histogram features of representative impedance profiles. Overlaid histogrambins on impedance profile (top) and tile bivariate histogram with cell count as color (bottom). × regions were excluded from the analysis as they correspond to reference bead population. (B) Correlation of 34D impedance features

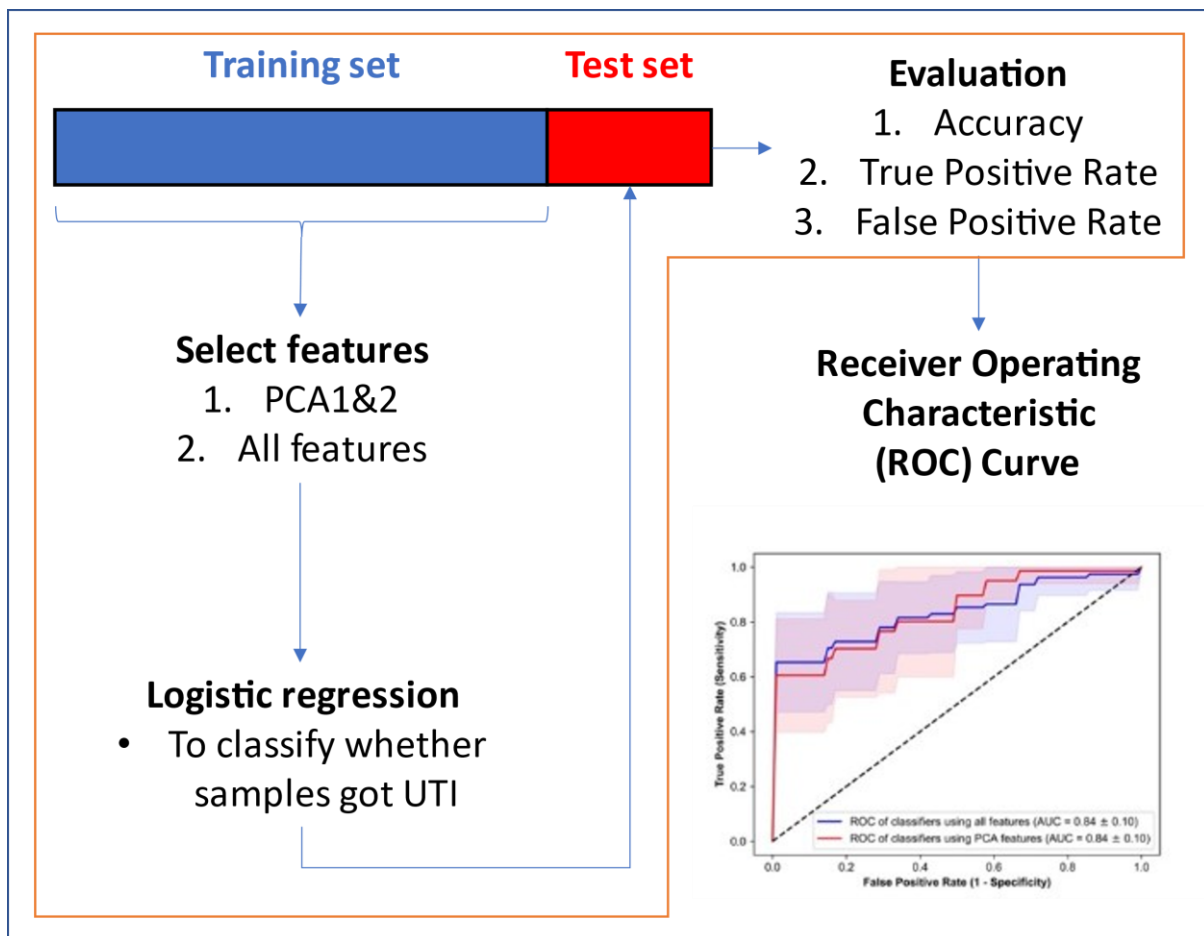


Figure S10 Validation of UTI classification model using repeated stratified 4-fold cross validation. For each data segments, data were shuffled and then used to train and evaluate model for 4 times (n repeat = 4). Following that, next data segments were randomly selected and repeated the same process again for 4 times (n fold = 4).

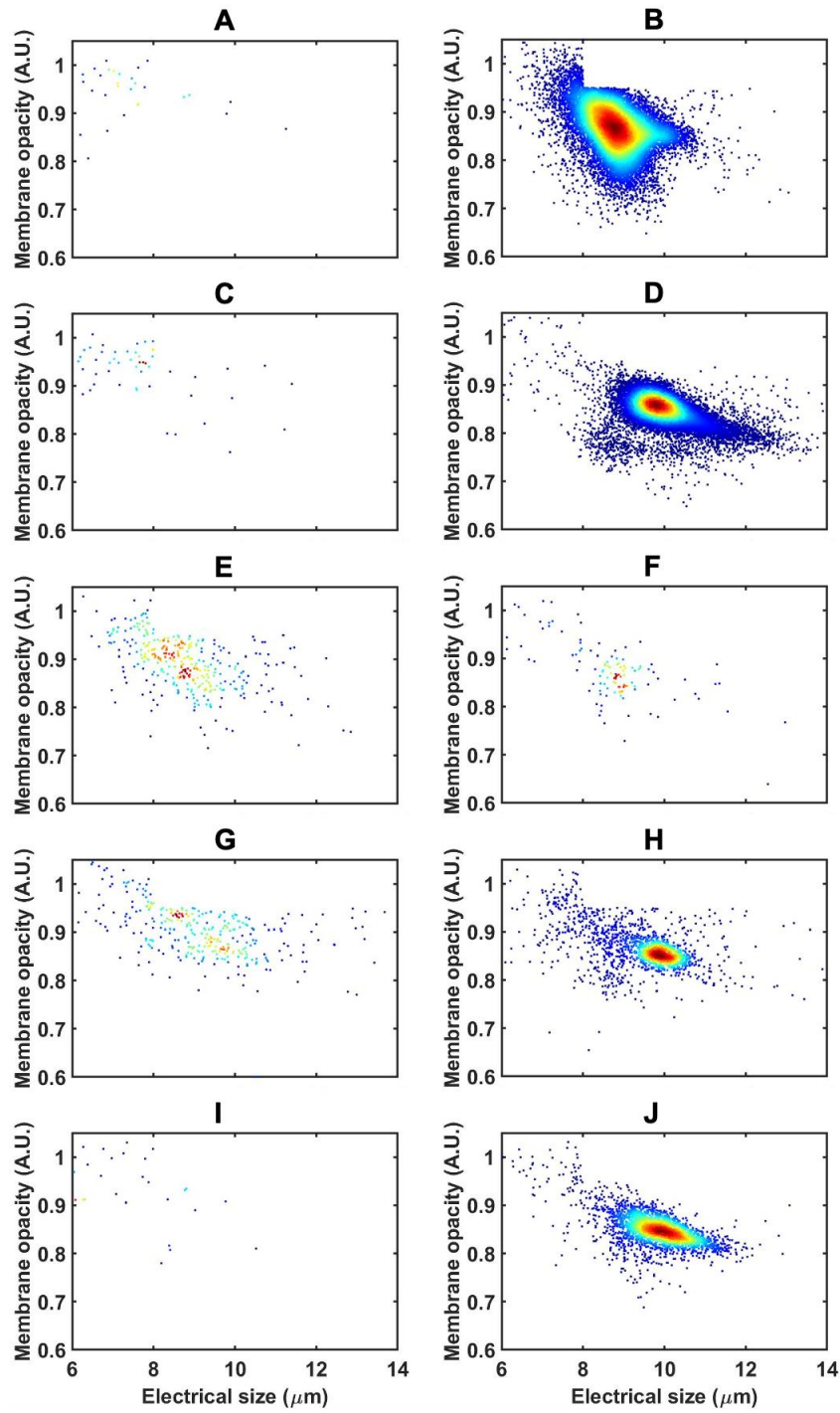


Figure S11 Urine impedance profiles from mock blinded experiments. Non-UTI (“no spiking”) urine samples are A, C, E, G, I, and UTI (“spiked”) urine samples are B, D, F, H, J.

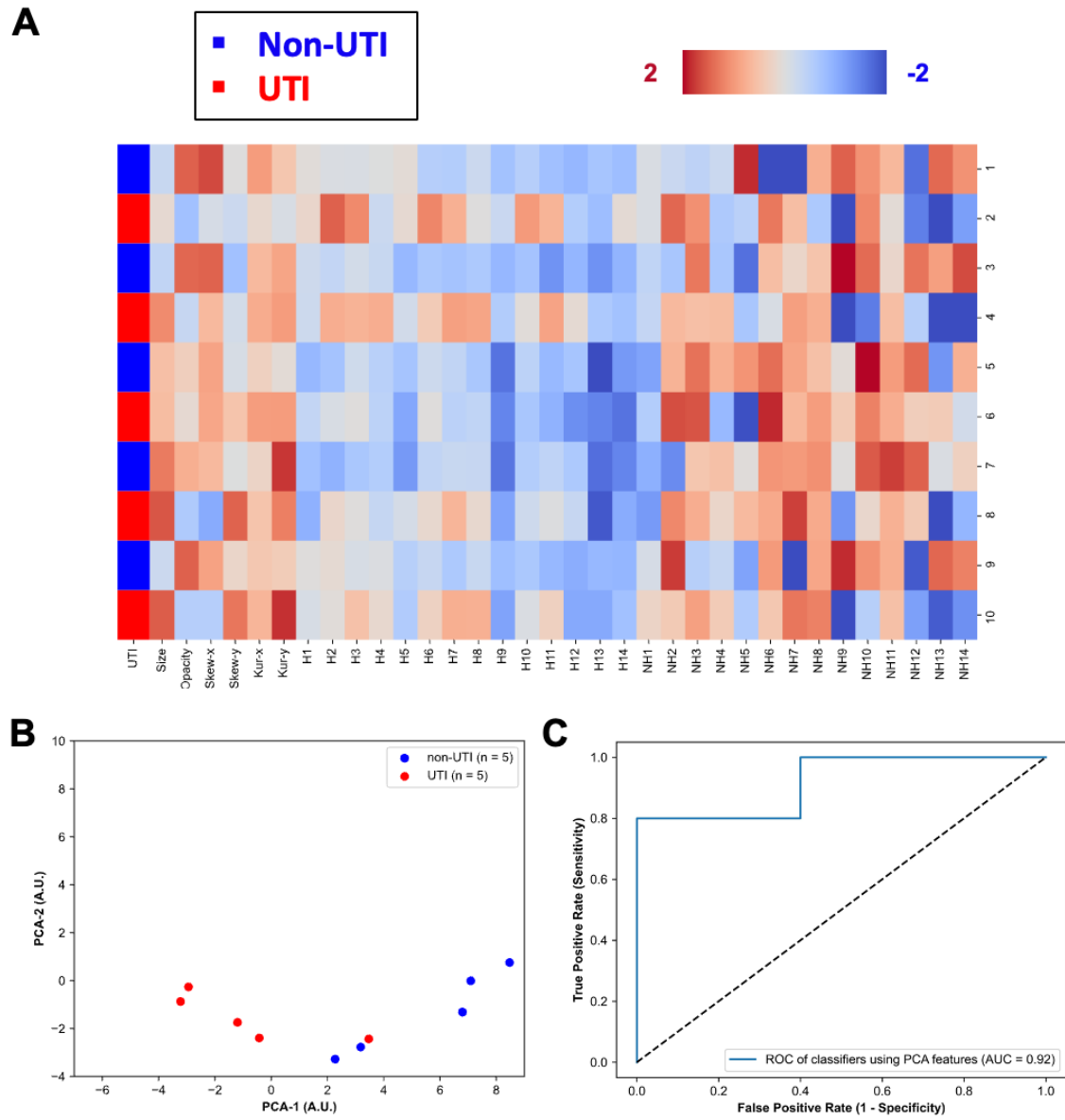


Figure S12 Urine analysis for mock blinded experiments. (A) Heatmap of urine sample characteristics. (B) PCA analysis. (C) ROC plot for UTI detection.